

6.1.5 Lesson Review

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Question 1

✓ Correct

An e-commerce company is expanding its AI capabilities to personalize customer recommendations. The AI team needs to experiment rapidly with various algorithms and deploy models efficiently.

This requires centralized access to powerful compute clusters, secure storage solutions, feature stores to manage engineered features, a model registry to track models and versions, and observability tools to monitor performance and usage—all of which need to be shared across multiple development teams.

Which role is primarily responsible for providing and maintaining this shared infrastructure that enables data scientists and engineers to experiment, deploy, and monitor AI models efficiently?

☒ Platform Engineer ✓ Correct

☐ AI Architect

☐ MLOps Engineer

☐ Data Scientist

Explanation

Platform Engineer is the correct answer. The Platform Engineer is responsible for delivering and maintaining the shared infrastructure such as compute clusters, secure storage, feature stores, model registries, and observability tools, ensuring AI teams can seamlessly experiment with and deploy models. This best matches the requirements of the scenario.

Data Scientist is incorrect. The Data Scientist is responsible for managing experiment infrastructure, provisioning storage, and maintaining feature stores, model registries, and observability tools to give the AI team full control over the end-to-end AI pipeline infrastructure. Although Data Scientists use these tools in their workflows, they do not provide or maintain them.

MLOps Engineer is incorrect. The MLOps Engineer owns all aspects of the AI platform including creating compute clusters, maintaining databases for storage and feature management, and building observability frameworks, while also developing machine learning models and designing experiments. While MLOps Engineers support pipeline automation, versioning, and model monitoring, and interact with infrastructure, their main focus is on automating processes and workflows for model operations.

AI Architect is incorrect. The AI Architect is tasked with designing infrastructure components, provisioning shared compute and storage resources, and maintaining feature stores, model registries, and monitoring products to ensure scalability and efficiency for AI development across the company. The AI Architect is responsible for outlining the high-level patterns and design of how components interact within the system.

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6.1.2 Important Roles in AI

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A financial institution has recently introduced AI to automate loan approvals. There have been several stakeholder concerns about potential bias, lack of oversight, and unclear decision trails.

As a member of the AI governance team, you must evaluate which procedural change would most effectively address these concerns and strengthen AI governance in line with best practices.

Which of the following procedural improvements should the organization prioritize?

- ☐ Launch a comprehensive awareness campaign for all business units about AI ethics, focusing on the importance of fairness, explainability, and incident reporting.
- ☐ Establish a quarterly committee meeting where representatives from legal, business, and IT discuss deployed AI systems, high-level incident trends, and new AI opportunities.
- ☒ Require that each AI model undergoes documented risk classification, design review, validation, regular monitoring, and includes a mandatory audit trail, with evidence retained for every decision made. ✔ Correct
- ☐ Increase the frequency of AI model retraining to incorporate the latest data trends and ensure model accuracy, allowing model teams to choose their documentation and review processes.

Explanation

Requiring that each AI model undergoes documented risk classification, design review, validation, regular monitoring, and includes a mandatory audit trail, with evidence retained for every decision made is the correct answer. This answer identifies a robust procedural approach, directly addressing bias (by requiring risk classification), oversight (documentation and review), and transparency (audit trail and retained evidence). It fully aligns with best practices ensuring processes are traceable and auditable, and that safeguards are applied consistently.

Increasing the frequency of AI model retraining to incorporate the latest data trends and ensuring model accuracy, allowing model teams to choose their documentation and review processes, is incorrect. While frequent retraining can mitigate bias due to shifting data, the absence of enforceable, standardized documentation and review procedures relies too heavily on individual teams' discretion, which can lead to inconsistent practices, untraceable decisions, and does not ensure that safeguards are consistently applied, as required for mature AI governance.

Establishing a quarterly committee meeting with representatives from legal, business, and IT is incorrect. Regular cross-functional meetings can support governance dialogue and foster ongoing oversight at a high level. Without clear, required procedures for documentation, review, and audit at every AI initiative stage, concerns about bias, transparency, and decision trails are not sufficiently addressed.

Launching a comprehensive awareness campaign for all business units is incorrect. Although awareness and training are important components of an AI governance program, this action alone is insufficient to address the structural concerns raised. It raises organizational consciousness but does not create or enforce procedural discipline. Problems with oversight, auditability, and transparency are likely to persist.

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6.1.1 Establish AI Governance

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Question 3

✓ Correct

A large healthcare provider is launching a new AI-powered diagnostic tool that processes sensitive patient data and generates automated health recommendations.

During threat modeling, the AI team identifies risks such as data leakage, adversarial attacks on model inputs, and unauthorized access to prompts and outputs.

The organization wants to analyze which expert is best positioned to integrate controls that protect data, isolate models, enforce access permissions, and implement adversarial testing to mitigate these threats.

Given the scenario, which AI role should take the lead in analyzing risks and designing comprehensive safeguards to secure the AI system throughout its lifecycle?

☒ AI Security Architect ✓ Correct

☐ Data Engineer

☐ AI Auditor

☐ MLOps Engineer

Explanation

AI Security Architect is the correct answer. The AI Security Architect is responsible for analyzing the system's security risks and designing controls to ensure data and model protection across the diagnostic tool's lifecycle. The AI Security Architect specifically focuses on designing and implementing security measures such as secrets management, access control, isolation, filtering, and adversarial testing. This role is dedicated to mitigating threats to the confidentiality, integrity, and safety of AI systems, and is best equipped for this type of analysis and safeguard design.

Data Engineer is incorrect. While Data Engineers are essential for ensuring that data pipelines are efficient and data quality is maintained, their expertise is not in the design and management of security controls, adversarial testing, or model isolation. These aspects require a specialized security focus.

MLOps Engineer is incorrect. The MLOps Engineer, through managing deployment automation, retraining workflows, and monitoring, is best positioned to identify threats and set up secrets management, model isolation, and adversarial testing to keep the AI-powered tool secure during operation. MLOps Engineers do support operational efficiency, versioning, and monitoring, but not security controls like threat analysis, secrets management, and adversarial testing.

AI Auditor is incorrect. The AI Auditor's main role is independent verification and assurance, not the direct analysis, design, or integration of security controls. While auditors can recognize gaps and audit compliance, they do not lead the implementation of security protections.

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6.1.2 Important Roles in AI

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An insurance organization recently adopted AI to automate claims processing. After a compliance audit, the company discovered gaps: several AI models had been deployed without the required approval evidence, records of data sources and model lineage were incomplete, and retention policies set in the company's AI procedures were not consistently enforced.

Leadership requests the AI team to identify the root causes and ensure these governance failures are prevented in the future by integrating policy requirements into day-to-day operations using automated workflows and monitoring systems.

Which team member should take the lead in analyzing these governance challenges and integrating controls to automate approval workflows, manage deployment risks, ensure evidence collection, and enforce data retention and audit requirements?

- ☐ Data Scientist
- ☐ AI Auditor
- ☐ Platform Engineer
- ☒ AI Governance Engineer ✓ Correct

Explanation

AI Governance Engineer is the correct answer. The AI Governance Engineer specializes in operationalizing policies by embedding approval workflows, collecting evidence automatically, and enforcing retention and audit requirements. This role ensures policies are applied consistently throughout the AI lifecycle.

Data Scientist is incorrect. While Data Scientists may document their work, their primary focus is on model development and analysis, not the automation and integration of governance controls and evidence collection as required for comprehensive policy compliance.

Platform Engineer is incorrect. Although Platform Engineers build infrastructure and may deploy monitoring tools, they do not integrate governance controls and automate approval workflows according to detailed policy requirements.

AI Auditor is incorrect. The AI Auditor is responsible for creating evidence collection systems and configuring policy-driven approval workflows directly within operational AI platforms to prevent governance issues. AI Auditors evaluate compliance and recommend improvements but do not design or embed automated governance processes or controls.

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6.1.2 Important Roles in AI

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Question 5

✓ Correct

Match each AI-related role on the left with its correct description on the right.

Builds and maintains data pipelines to acquire, clean, label, and deliver quality-controlled data for AI initiatives.

✓ Data Engineer

Explores and analyzes data to define problems, select features, and develop baseline AI models that show value.

✓ Data Scientist

Transforms prototype models into production-grade systems by optimizing code, packaging dependencies, and constructing services for reliable, real-world operation.

✓ Machine Learning Engineer

Designs safeguards to protect AI systems from threats such as data leakage, model poisoning, and prompt injection, ensuring that data, models, and prompts are secure.

✓ AI Security Architect

Provides independent assurance by reviewing artifacts and logs, verifying policy adherence, and confirming that monitoring and incident handling are operating as intended.

✓ AI Auditor

Explanation

The following are the correct description matches to the listed AI-related roles:

1. **Data Engineer.** The data engineer's primary responsibility is to build and maintain the complex pipelines that gather, prepare, and ensure the quality of data used in AI systems. This foundation is crucial for trustworthy and reliable AI modeling.
2. **Data Scientist.** The data scientist explores the data provided, determines the relevant features, clarifies problems, and crafts initial or baseline AI models—these early models demonstrate the possible value of more robust solutions.
3. **Machine Learning Engineer.** Machine learning engineers take the experimental models created by data scientists and turn them into robust, scalable products. This involves code optimization, dependency management, and designing production-ready inference services.
4. **AI Security Architect.** This role is centered on risk and protection. The AI security architect devises and implements measures to secure data, models, and processes from a range of possible attacks and vulnerabilities using technical and procedural safeguards.
5. **AI Auditor.** The AI auditor ensures the whole governance process works as intended—reviewing artifacts, logs, and documentation to verify that governance policies are consistently followed, and that incident response and monitoring are functioning as required.

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A mid-sized healthcare organization is rolling out a new AI-powered diagnostic tool across several medical departments. Leadership is concerned about maintaining oversight and regulatory compliance as the system uses sensitive patient data and learns from real-time inputs.

As part of establishing AI governance, your team is tasked with recommending technical guardrails that ensure secure and compliant AI operations.

Which of the following actions BEST exemplifies applying technical guardrails to support AI governance in this scenario?

- ☐ Conducting periodic staff training to raise awareness about ethical AI use and publishing a general AI policy document emphasizing patient data privacy.
- ☐ Using anonymized datasets during model development and reviewing the AI system after incidents, but allowing developers open access to production data for troubleshooting.
- ☒ Implementing access controls that limit model deployment and sensitive data access to authorized personnel, along with automated logging and monitoring of every model inference and data transaction. ✓ Correct
- ☐ Appointing a joint review committee from IT and compliance to sign off on new AI features before release, focusing on up-front risk approval.

Explanation

Implementing access controls that limit model deployment and sensitive data access to authorized personnel, along with automated logging and monitoring of every model inference and data transaction is the correct answer. This option directly implements technical guardrails: access controls based on least privilege, logging and audit trails, and real-time monitoring are explicit, technical measures embedded in the platform to enforce policy, ensure accountability, and maintain visibility over AI use. These measures support compliance, security, and the organization's governance framework.

Conducting periodic staff training to raise awareness about ethical AI use and publishing a general AI policy document is incorrect. They are not technical guardrails. Technical guardrails are programmatic, system-level controls that enforce governance automatically (such as access control, logging, and data masking). Policies and training support governance but do not directly enforce or monitor AI activity in the operational environment.

Appointing a joint review committee from IT and compliance to sign off on new AI features before release, focusing on up-front risk approval is incorrect. These not a technical guardrail. While important for governance, this approach primarily provides human review and may ensure policies are considered. However, it does not automatically and continuously enforce compliance or provide monitoring capabilities in the deployment and operation of AI systems.

Using anonymized datasets during model development and reviewing the AI system after incidents, but allowing developers open access to production data is incorrect. Allowing open developer access to production data contradicts technical guardrail principles such as "least privilege" and strict access controls. In fact, this approach increases the risk of data breaches, unauthorized changes, and reduces accountability. Further, it demonstrates a reactive rather than proactive control model.

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Question 7

✓ Correct

A financial services company is piloting an AI-driven fraud detection application. After the Data Engineer supplies a cleaned and labeled dataset, the Data Scientist identifies relevant features and creates a promising prototype model that shows strong results in initial tests.

The company now needs to deploy this model for real-time monitoring of transactions. This deployment requires the model code to be optimized, dependencies to be packaged, an inference service to be developed, and evaluation tests that ensure resilience under real-world traffic.

Which AI team member's primary responsibilities best fit these deployment tasks?

☒ Machine Learning Engineer ✓ Correct

☐ AI Security Architect

☐ Data Engineer

☐ MLOps Engineer

Explanation

Machine Learning Engineer is the correct answer. The Machine Learning Engineer takes the prototype model from the Data Scientist, optimizes it for production, packages any dependencies, develops inference services, and creates robust evaluation tests to ensure the model performs reliably at scale. The responsibilities of the Machine Learning Engineer include converting prototypes to production-ready models by optimizing code, packaging dependencies, creating inference services, and designing evaluation tests.

AI Security Architect is incorrect. The AI Security Architect is tasked with reviewing model deployments, designing secrets management, and additionally takes the lead on optimizing code and setting up inference services to streamline deployment and performance for production models. The AI Security Architect's main role is to implement safeguards for data, models, and prompts (such as secrets management and access control).

Data Engineer is incorrect. The Data Engineer manages the deployment by overseeing the development of inference services, packaging dependencies, and establishing model evaluation tests, in addition to maintaining data pipelines. While the Data Engineer is crucial for preparing data and maintaining data pipelines, their responsibilities do not include model code optimization or deployment tasks.

MLOps Engineer is incorrect. The MLOps Engineer directly receives prototype models from the Data Scientist, optimizes them for production, packages dependencies, and creates model evaluation tests before deployment to the production environment. The MLOps Engineers focus on automating the model lifecycle, setting up CI/CD pipelines, and monitoring models.

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6.1.2 Important Roles in AI

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A financial services firm has just implemented its first AI-powered customer support chatbot. The leadership team wants to make sure that their AI governance, risk, and compliance (GRC) program matures appropriately over time.

They ask you how to progress from an initial stage of AI GRC maturity to a more advanced stage in line with best practices.

Which of the following steps BEST demonstrates advancing the firm's AI GRC maturity according to the AI GRC Maturity Roadmap?

- ☐ Immediately roll out complex AI monitoring systems organization-wide before establishing basic policies or conducting an inventory of existing AI use cases.
- ☐ Allow business units to publish their own AI guidelines and requirements without oversight, encouraging flexibility in AI adoption.
- ☒ Begin automating evidence collection for regulatory compliance and arrange for independent (third-party) validation of high-risk AI systems. ✓ Correct
- ☐ Focus solely on conducting quarterly employee workshops about digital ethics and responsible AI to increase staff awareness.

Explanation

Beginning the automation of evidence collection for regulatory compliance and arranging for independent (third-party) validation of high-risk AI systems is the correct answer. This option matches the advanced stages in the AI GRC Maturity Roadmap. After initial steps like publishing policies and inventorying AI usage, organizations should automate evidence collection and obtain independent validation of high-risk systems as part of maturing their AI GRC program.

Focusing solely on conducting quarterly employee workshops about digital ethics and responsible AI to increase staff awareness is incorrect. This answer does not represent an advancement along the AI GRC maturity roadmap. The roadmap calls for maturing technical, operational, and compliance controls (like evidence automation and validation), not just training and awareness initiatives. Relying only on educational workshops would leave significant gaps in systematic risk management and technical compliance for AI maturity.

Allowing business units to publish their own AI guidelines and requirements without oversight, encouraging flexibility in AI adoption, is incorrect. Allowing each unit to independently set requirements without broader oversight can create inconsistencies, make compliance difficult to enforce, and increase the risk of "shadow AI"—solutions developed without proper risk assessment or alignment to organizational standards. This step would actually hinder, rather than advance, the maturity of the AI GRC program and could expose the firm to increased legal, ethical, and operational risks.

Immediately rolling out complex AI monitoring systems organization-wide before establishing basic policies or conducting an inventory of existing AI use cases is incorrect. This contradicts the maturity progression outlined in the roadmap. According to best practices, organizations should start with concise policies and an inventory, then move forward to implement registries, monitoring, and automated controls.

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Which statement BEST explains the main function of an AI policy within an organization's AI governance framework?

- ☐ An AI policy is a collection of software tools for monitoring AI models during production.
- ☐ An AI policy outlines which AI-related roles will be offered the highest salaries in the company.
- ☐ An AI policy consists solely of technical details about model training algorithms.
- ☒ An AI policy specifies what is permitted with AI systems, what evidence is needed for audits, and who is authorized to make key decisions. ✓ Correct

Explanation

An AI policy specifies what is permitted with AI systems, what evidence is needed for audits, and who is authorized to make key decisions is the correct answer. An AI policy clarifies what is allowed, details the required evidence for accountability and audits, and establishes who is authorized to make decisions about AI. These elements are central to the organization's oversight and responsible management of AI initiatives.

An AI policy outlines which AI-related roles will be offered the highest salaries in the company is incorrect. Salary decisions for AI-related roles may be addressed by human resources policies, not by the AI policy. The AI policy's main function is about defining permissions, required evidence, and authority related to AI systems, not compensation.

An AI policy is a collection of software tools for monitoring AI models during production is incorrect. While monitoring tools support the enforcement of policy, the policy itself is a formal document, not a set of technical tools. The policy establishes the rules that such tools help to enforce.

An AI policy consists solely of technical details about model training algorithms is incorrect. Although some technical aspects may be referenced, an AI policy is broader and includes ethical guidelines, data handling requirements, decision-making authorities, and compliance standards—not just algorithm details.

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 6.1.1 Establish AI Governance

Question 10

✓ Correct

Which of the following are common ways to position an AI Center of Excellence (CoE) within an organization's AI governance structure? (Select three.)

- ☐ Outsourced-only model
- ☐ Project-based temporary committee
- ☒ Centralized model ✓ Correct
- ☒ Federated model ✓ Correct
- ☐ Matrix reporting model between HR and Finance
- ☐ Decentralized vendor partnership
- ☒ Hybrid model ✓ Correct

Explanation

Centralized model is a correct answer. The centralized model places most AI development and all governance within one team, helping to speed up standardization. This is one of the three common ways for positioning an AI CoE.

Federated model is a correct answer. The federated model distributes AI developers across business units while the CoE provides standards and assurance. This is one of the three common ways for positioning an AI CoE.

Hybrid model is a correct answer. The hybrid model centralizes platforms, policies, and reviews within the CoE, but allows business units some autonomy within defined boundaries. This model balances control and efficiency and is frequently used.

Outsourced-only model is incorrect. An outsourced-only model, where all AI responsibilities are handed off to an external organization, is not identified as a common way to structure an internal CoE. Internal cross-functional collaboration is a core feature of a CoE.

Decentralized vendor partnership is incorrect. While vendor partnerships can support AI efforts, a decentralized vendor partnership is not recognized as one of the three common CoE positions. The focus is on internal organizational structures rather than external partnerships.

Matrix reporting model between HR and Finance is incorrect. A matrix reporting structure between HR and Finance does not pertain to AI governance or CoE placement. It is unrelated to the context and functions as an AI CoE.

Project-based temporary committee is incorrect. A project-based temporary committee is different from a CoE, which is a more permanent, cross-functional structure designed to standardize and guide AI initiatives over time, not just for single projects.

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6.1.1 Establish AI Governance

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